Problem Set 1

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Due: October 9, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 9, 2025. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

1. Find a 90% confidence interval for the average student IQ in the school.

```r
sample_mean <- mean(y)
sample_sd <- sd(y)
t90 <- qt(1 -( 1 - .90)/2, df = 24)
n <- length(y)
lower_90 <- sample_mean - (t90 * (sample_sd/sqrt(n)))
upper_90 <- sample_mean + (t90 * (sample_sd/sqrt(n)))
confint90 <- c(lower_90, upper_90)
```

We can infer this by stating that if we repeated the sampling process 100 times, 90 of the intervals would contain the true population mean, which would be between the bounds of 93.96 and 102.92.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

   Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```r
alpha <- 0.05
t_stat <- (sample_mean - 100) /(sample_sd/ sqrt(n))
df <- n - 1
p_value_upper <- pt(-abs(t_stat), df, lower.tail = FALSE)
ifelse(p_value_upper > alpha, "We failed to reject the Null Hypothesis",
    "We reject the Null Hypothesis")
"We failed to reject the Null Hypothesis"
```

   The Null Hypothesis: The schools IQ score is less than or equal to 100. $H_0 : \mu \leq 100$

   Alternative Hypothesis: The mean of the schools IQ is greater than 100. $H_a : \mu > 100$

   Since our P-value is 0.7 which is greater than 0.05 we have failed to reject the Null Hypothesis that the school's IQ test scores are less than or equal to 100.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

-

```r
X1_Y <- ggplot(expenditures, aes(x = X1, y = Y)) +
        geom_point(alpha = 0.7, color = "blue") +
        geom_smooth(method = "lm", se = TRUE, color = "red") +
        labs(x = "Per Capita Personal Income", y = "Housing Assist. Expenditure") +
        theme_minimal()

X2_Y <- ggplot(expenditures, aes(x = X2, y = Y)) +
        geom_point(alpha = 0.7, color = "blue") +
        geom_smooth(method = "lm", se = TRUE, color = "red") +
        labs(x = "Financial Insecurity", y = "Housing Assist. Expenditure") +
        theme_minimal()

X3_Y <- ggplot(expenditures, aes(x = X3, y = Y)) +
        geom_point(alpha = 0.7, color = "blue") +
        geom_smooth(method = "lm", se = TRUE, color = "red") +
        labs(x = "Residing in Urban Areas", y = "Housing Assist. Expenditure") +
        theme_minimal()

X1_2 <- ggplot(expenditures, aes(x = X2, y = X1)) +
  geom_point(alpha = 0.7, color = "blue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
```

```r
  labs(x = "Income Per Capita", y = "Reported Financial Insecurity") +
  theme_minimal()

X2_3 <- ggplot(expenditures, aes(x = X2, y = X3)) +
                    geom_point(alpha = 0.7, color = "blue") +
                    geom_smooth(method = "lm", se = TRUE, color = "red") +
                    labs(x = "Reported Financial Insecurity", y = "Residents in Urban Areas") +
          theme_minimal()
X1_3 <- ggplot(expenditures, aes(x = X1, y = X3)) +
                        geom_point(alpha = 0.7, color = "blue") +
                        geom_smooth(method = "lm", se = TRUE, color = "red") +
                        labs(x = "Income Per Capita", y = "Residents in Urban Areas") +
               theme_minimal()
combined_plot <- grid.arrange(X1_Y, X2_Y, X3_Y, X1_2, X1_3, X2_3, ncol = 3, nrow = 2 )
```

X1, Y - There is a somewhat weak linear relationship between housing assistance expenditure and per capita personal income, with significant outliers. Overall, the graph indicates that states with higher per capita personal income tend to have a higher rate of expenditure on housing and shelter assistance. However, it is essential to note that some outlier states with higher per capita personal income spend a significantly lower amount on housing assistance. In contrast, other states spend a lot more than the average.

X2, Y - This graph indicates a nonlinear relationship between housing assistance expenditure and reported financial insecurity. States that have both the highest and lowest numbers of reported financial insecurity have some of the highest spending on housing assistance.

X3, Y - There is a weak linear relationship between housing assistance expenditure and the number of people residing in urban areas. There are notable outliers, demonstrating that some areas with a high concentration of people living in urban areas have relatively lower expenditures on housing assistance.

X1, X2 - There is no observable visual correlation between reported financial insecurity and income per capita. Among those with the highest rates of income per capita, both the highest and lowest rates of financial insecurity are also reported.

X1, X3 - There is a somewhat weak linear relationship between Income Per Capita and the number of residents in urban areas. The graph indicates as income per capita increases so does the number of residents in urban areas. However, there are significant outliers, where states with high income per capita have a low proportion of people residing in urban areas.

X3, X2. Looking at this graph, there is no clear visual indication of a relationship between the two variables. Those who have reported both the highest and lowest

amounts of economic insecurity reside in states with a high proportion of the population living in urban areas.

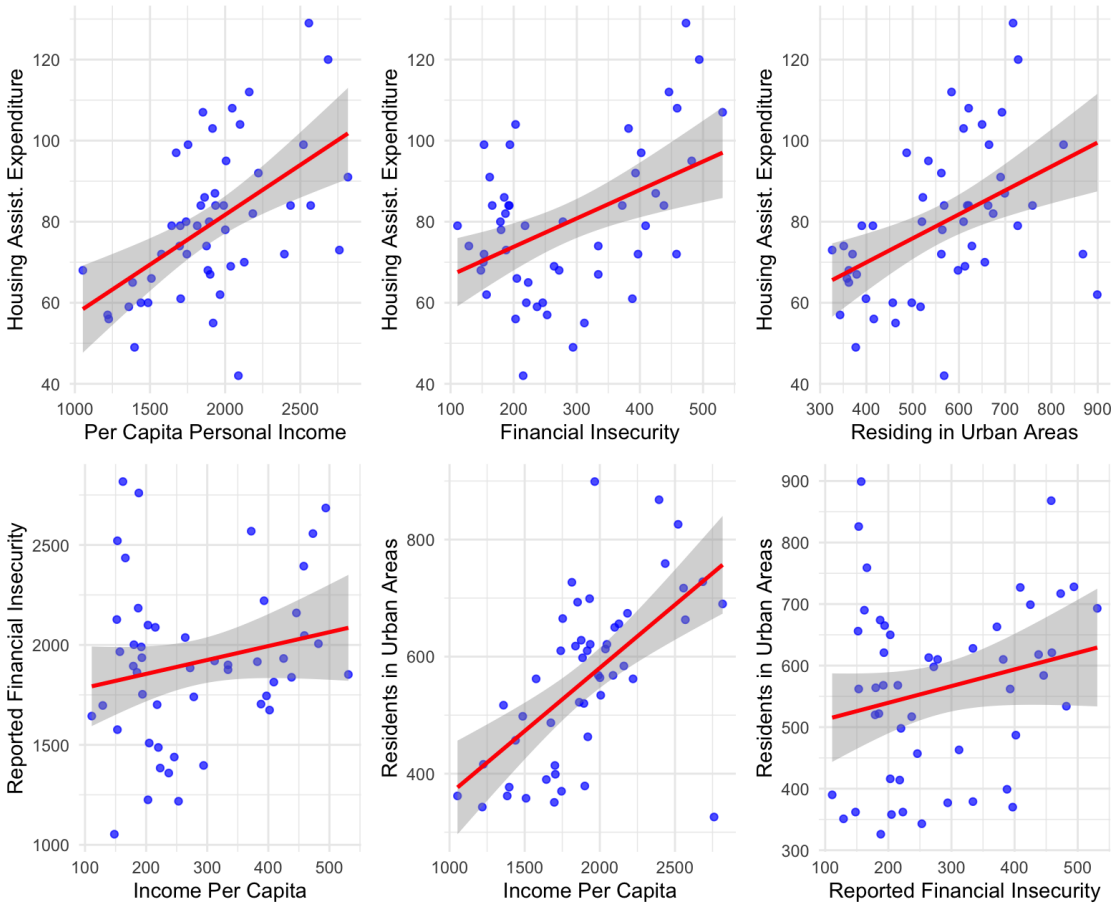

Figure 1: Relationships among Y, X1, X2, X3

- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```
#Code for Question 2, Part 2
Region_Fac <- as.factor(expenditures$Region)

ggplot(expenditures, aes(X1, Y, group = Region_Fac)) +
        geom_point() +
        scale_colour_viridis_d()
theme_minimal()
```

On average the West Coast has the highest expenditure on housing assistance per capita.

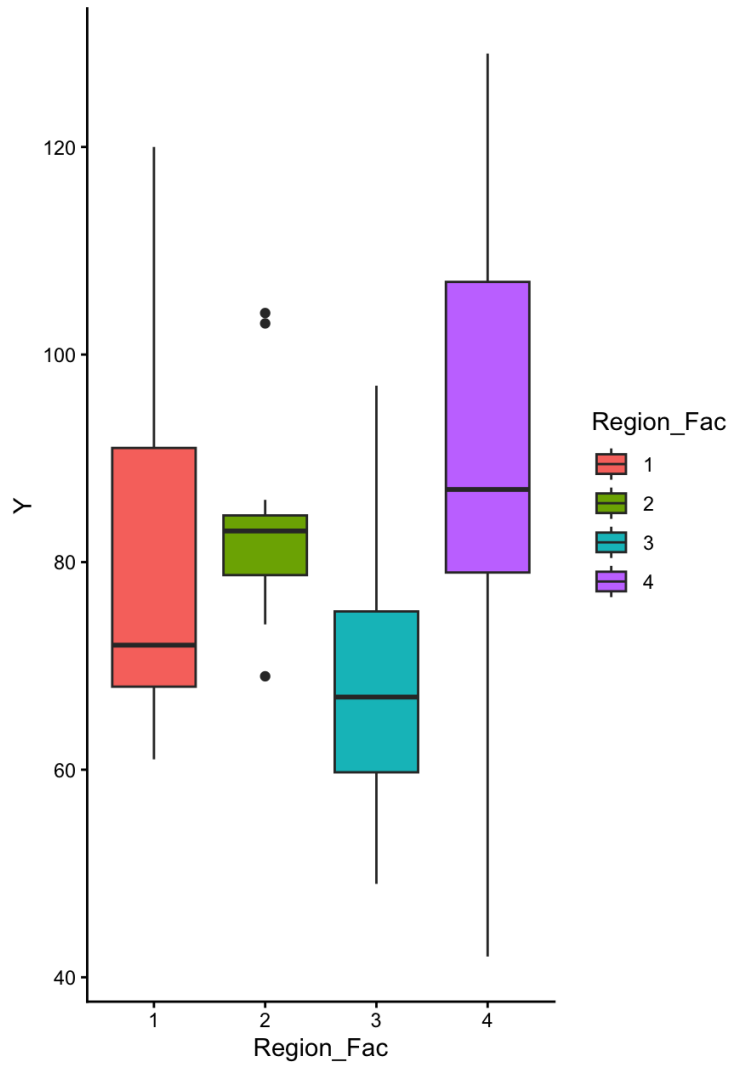

Figure 2: Housing/Shelter Assistance Expenditure and Region

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```
#Code for Question 2, Part 3
ggplot(expenditures, aes(X1, Y, group=Region_Fac)) +
        geom_point(aes(shape = Region_Fac, color = Region_Fac)) +
        theme_classic() +
       scale_color_discrete(
         name = "Regions",
         labels = c("Northeast", "North Central", "South", "West")
       ) +
       scale_shape_manual(
         name = "Regions",
```

```
        values=c(1,3,9,13),
        labels = c("Northeast", "North Central", "South", "West")
      ) +
       theme(
             legend.position = "top",
             axis.line = element_line(linewidth = 0.5),
             axis.line.x.bottom = element_line(colour = "Black"))
```

- This graph shows us the weak linear relationship between housing assistance expenditure and per capita personal income. Having this graph shaped by region allows us to identify that the region spending the least on housing assistance is the South. This is indicated by the concentration of blue plot points in the bottom left of the graph. However, it is challenging to determine which region has the highest expenditure on housing assistance solely from the plot.

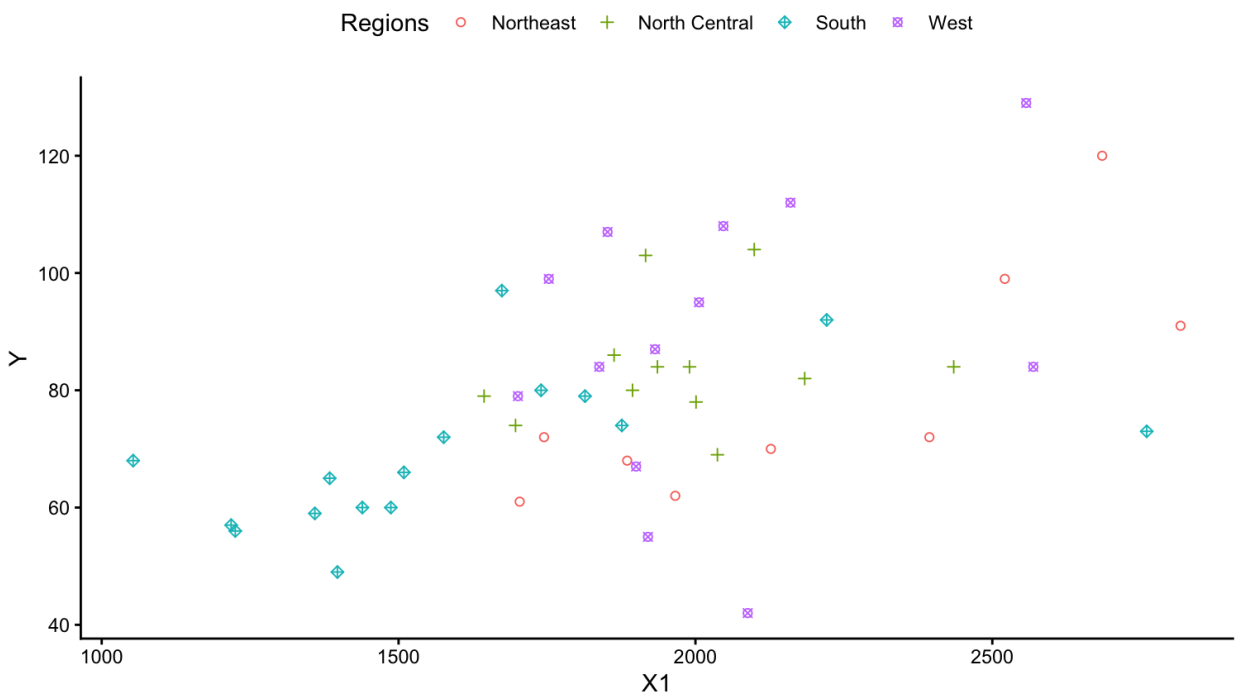

Figure 3: Y and X1 grouped by Region